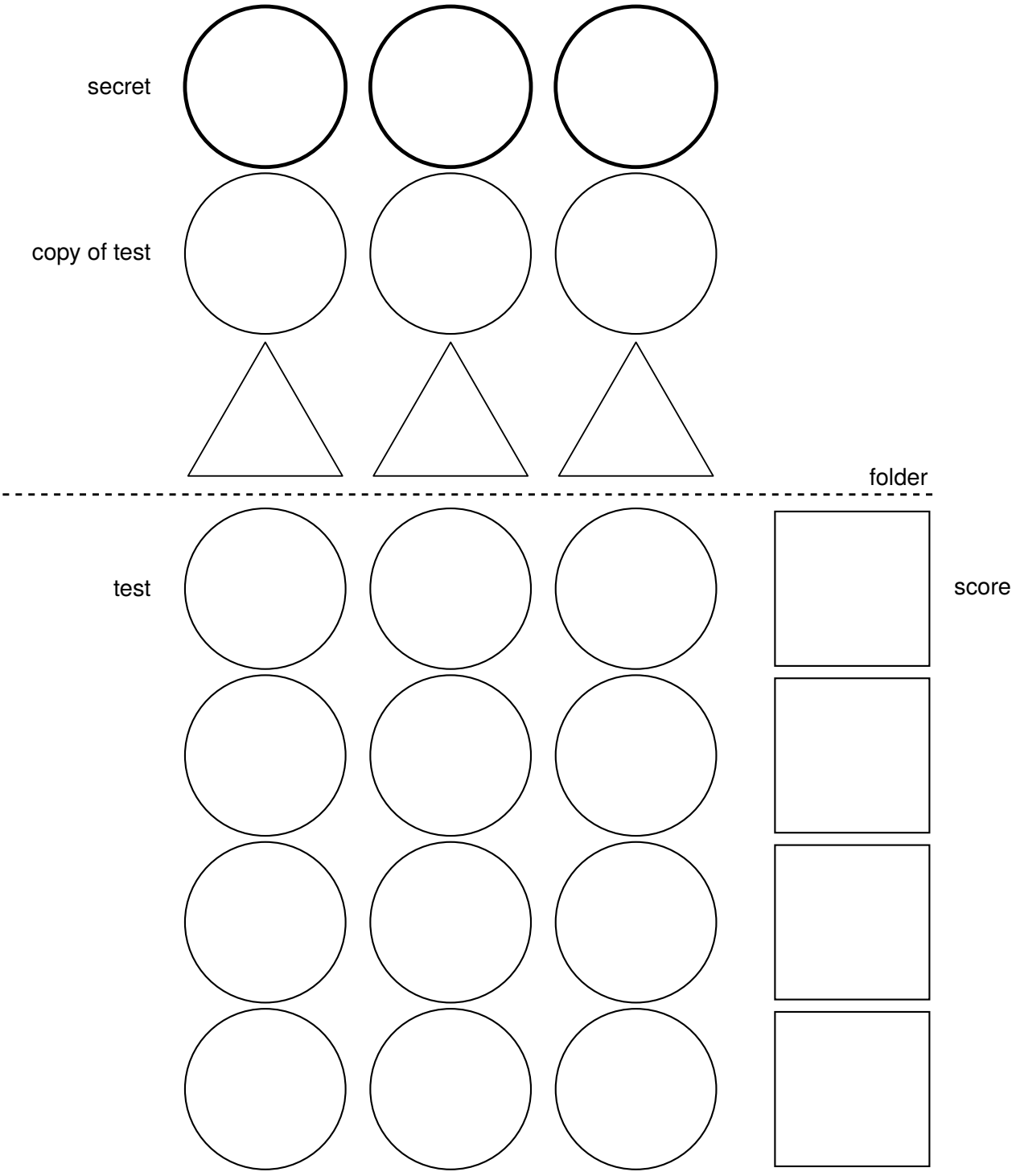
Problem 1: Your partner thinks of a whole number from 1 to 16. Ask yes-or-no questions until you are sure of the number. What is the smallest number of questions that always works? Explain why fewer questions cannot always work. Then answer the same questions for a number from 1 to 20.

The keeper hides a red/yellow secret row. The breaker fills a test row with red/yellow counters. Behind the folder, the keeper copies the test below the secret and puts a green triangle under each matching place. The breaker records the number of triangles as the score. After a final guess, uncover the secret and check the scores.

Problem 2: Play the code game with three counters, taking turns. Find each secret with as few tests as you can.



Problem 3: For each of these games, find every secret that fits all the scores. If no secret fits, explain how that could happen.

			score
R	R	Y	2
R	Y	R	2
Y	R	R	2

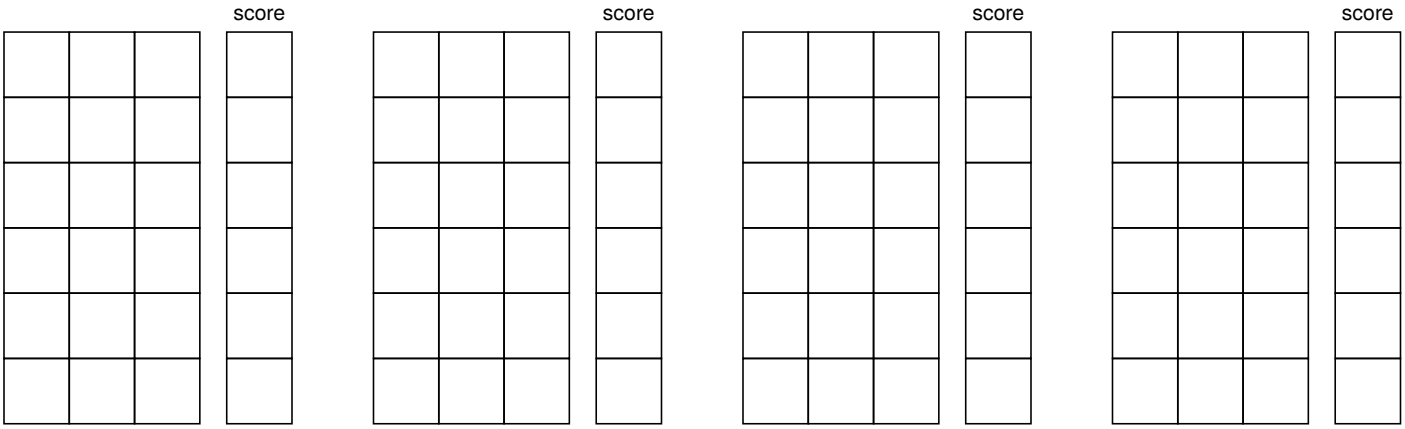
			score
R	R	R	2
R	Y	Y	2

				score
R	R	Y	Y	3
Y	R	Y	R	1
R	R	R	Y	2

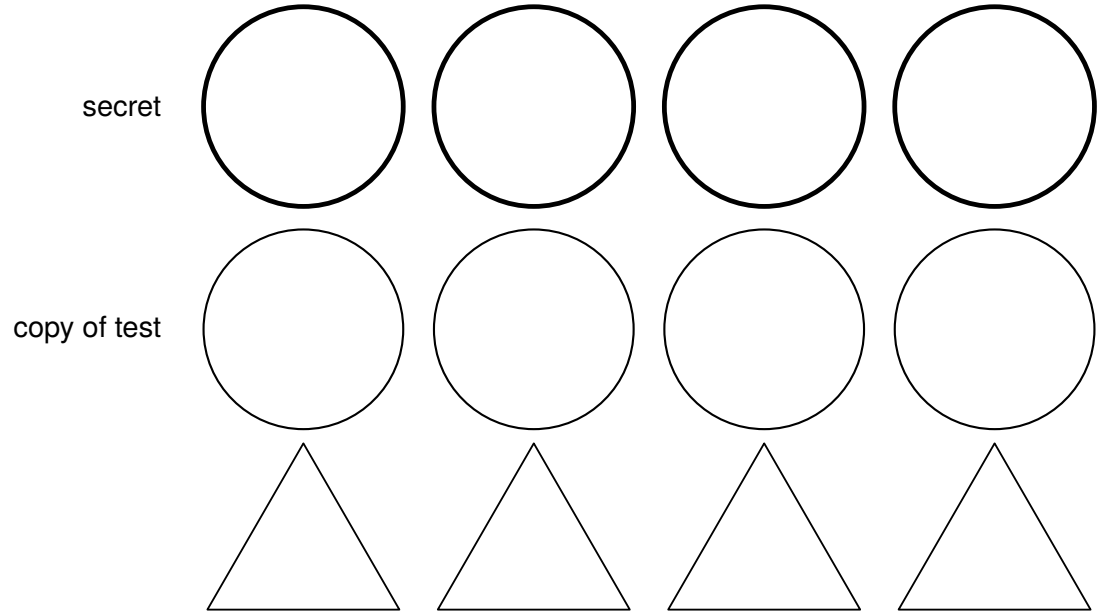
				score
R	R	Y	Y	2
R	Y	R	Y	2
R	Y	Y	R	2

				score
R	Y	R	Y	1
R	R	Y	Y	3
Y	Y	Y	Y	2

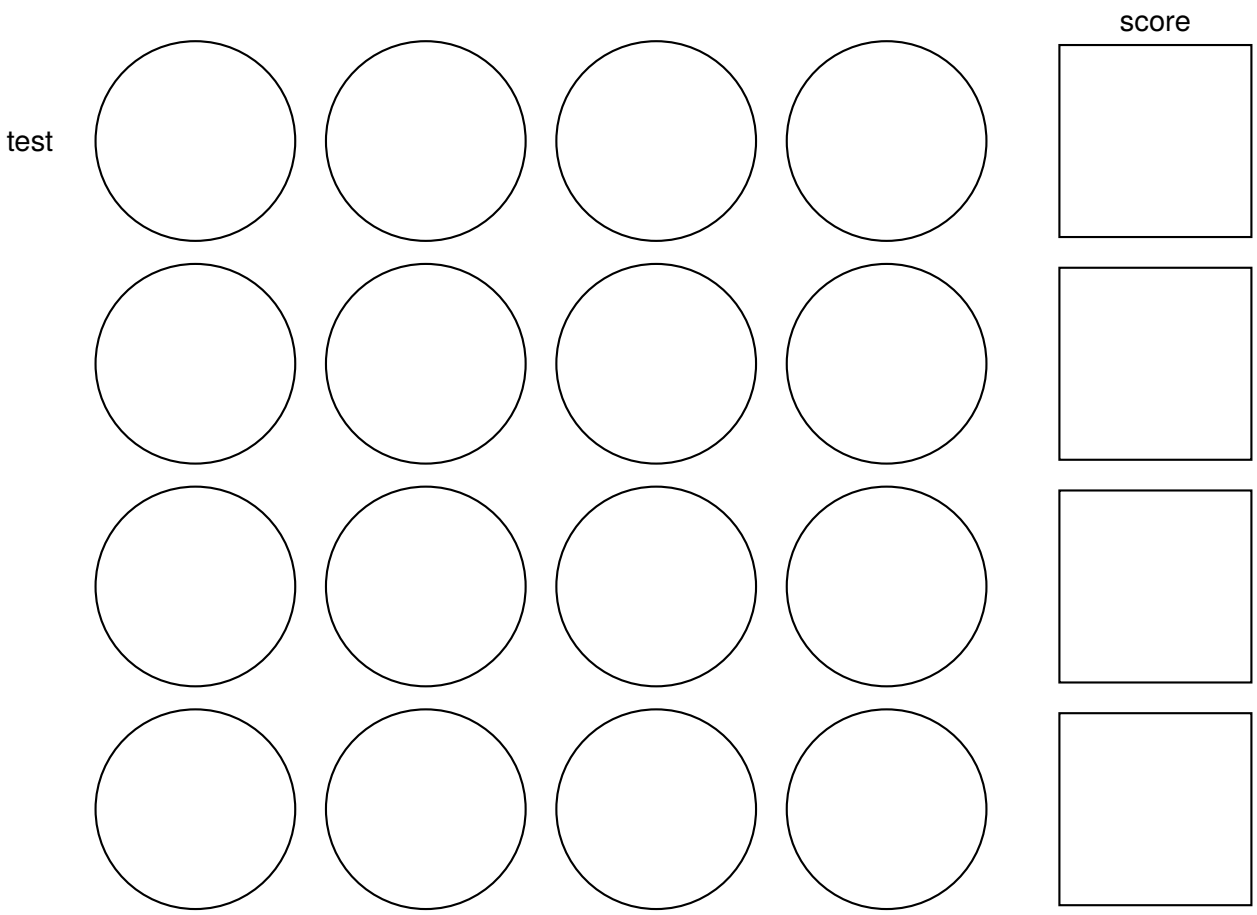
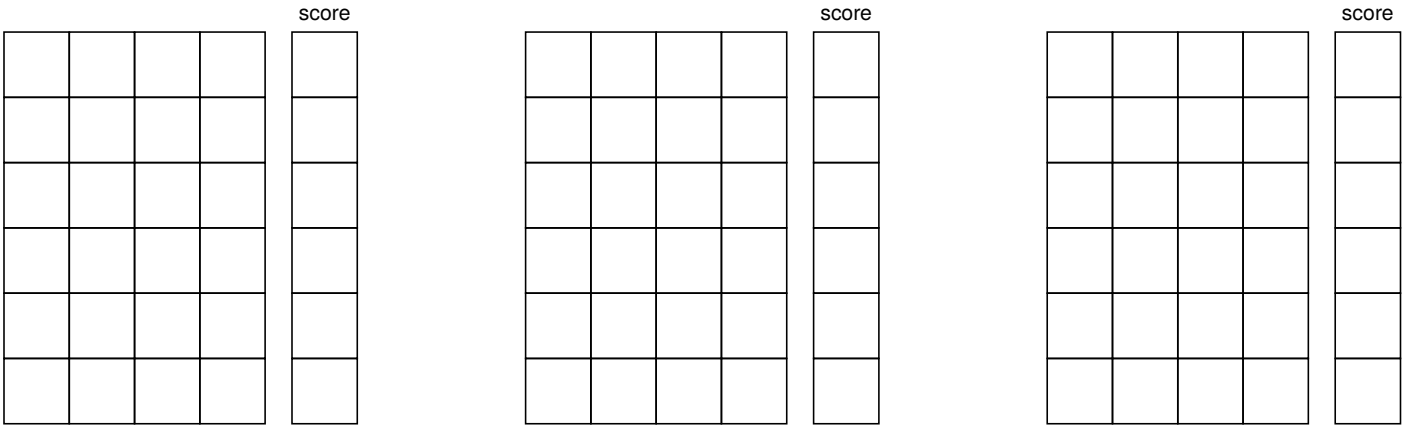
Problem 4: Find a way to choose your tests so that you always know a three-counter secret after three tests or fewer. Describe it so that someone else could use it.



Problem 5: Play the code game with four counters, taking turns. Find each secret with as few tests as you can.



folder



Problem 6: Find a way to choose your tests so that you always know a four-counter secret after four tests or fewer. Describe it so that someone else could use it.